

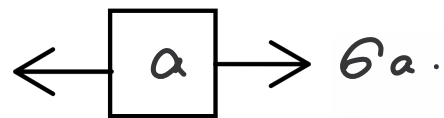
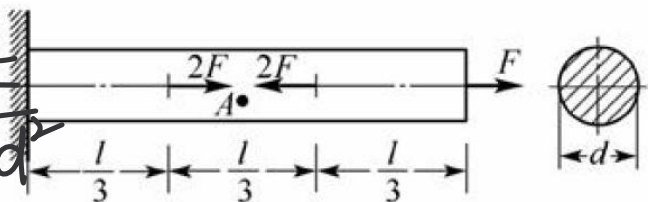
17讲-解析法-作业1:

如图所示4个构件。每个构件中A点和B点处取出单元体，标明单元体各面上的应力。图中未标明单位的长度量，单位均为mm。

$$F - 2F - F_A = 0$$

$$F_A = -F$$

$$G = \frac{F_A}{A} = \frac{-4F}{\pi d^2}$$



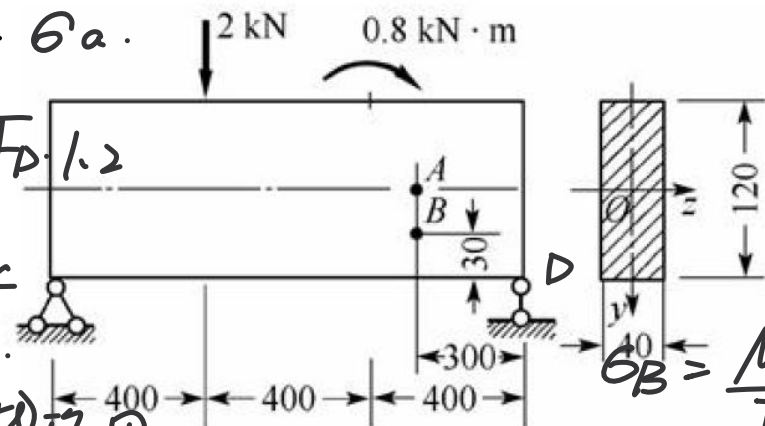
$$2 \times 0.4 + 0.8 = F_D \cdot 1.2$$

$$\therefore F_D = \frac{4}{3} \text{ kN}$$

$\therefore A$ 在中性面上。

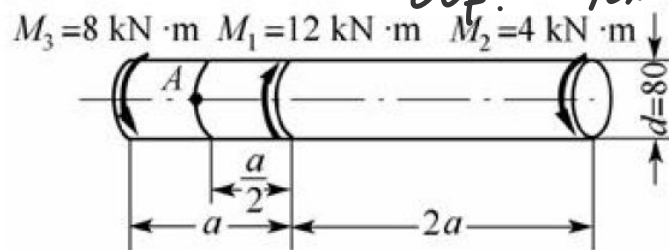
$\therefore A$ 处是纯弯曲作用。

$$\tau_A = \frac{3}{2} \frac{F_s}{A} = \frac{3}{2} \times \frac{\frac{4}{3} \times 10^3}{0.12 \times 0.04} = 0.42 \text{ MPa}$$



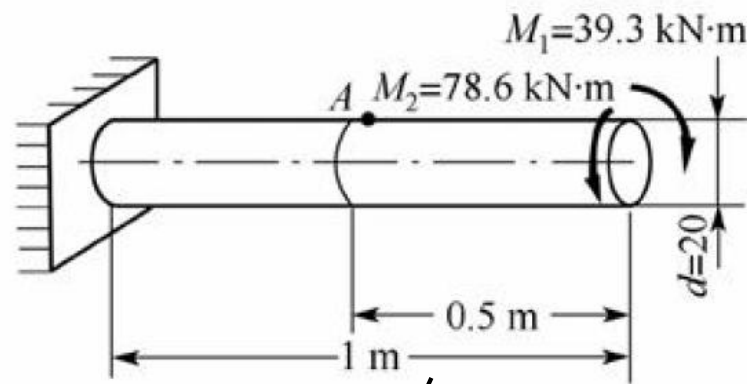
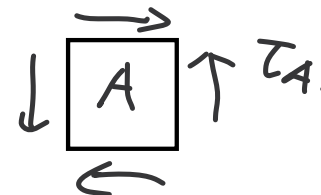
$$G_B = \frac{M_y}{I_z}$$

$$= \frac{\frac{4}{3} \times 10^3 \times 0.3 \times 0.02}{0.04 \times 12^3 / 12} = 2.1 \text{ MPa}$$



$$M_A = M_3 = 8 \text{ kN} \cdot \text{m} \quad F_A = 0$$

$$\tau_A = \frac{M_A}{W_p} = \frac{8 \times 10^3}{\pi \times 0.08^3 / 16} \text{ Pa} = 79.6 \text{ MPa}$$

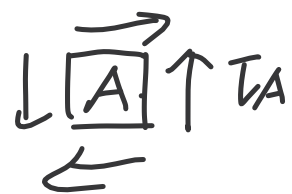


$$\tau_A = \frac{M_2}{W_p} = \frac{78.6 \times 10^3}{\pi \times 0.02^3}$$

$$= 50 \text{ MPa}$$

$$G_A = \frac{M_1}{W_z} = \frac{39.3 \times 10^3}{\pi \times 0.02^3} = 50 \text{ MPa}$$





17讲-解析法-作业2:

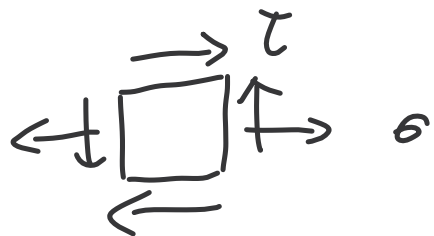
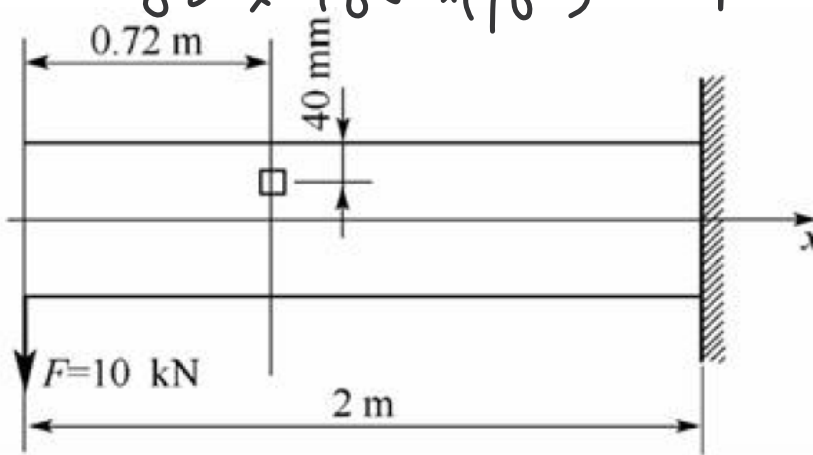
$$\tau_B = \frac{F_S}{2I_z} \left(\frac{h^2}{4} - y^2 \right) = \frac{\frac{10}{3} \times 10^3}{2 \times \frac{0.04 \times 0.12^3}{12}} \times \left(\frac{0.12^2}{4} - 0.03^2 \right) = 0.31 \text{ MPa}$$



试用解析法求图示悬臂梁距离自由端为0.72m的截面上，在顶面以下40mm的一点处的最大及最小主应力，并求最大主应力与x轴之间的夹角。

$$\sigma = \frac{M y}{I_z} = \frac{12 M y}{b h^3} = \frac{12 \times 0.72 \times 10 \times 10^3 \times 40 \times 10^{-3}}{80 \times 10^{-3} \times (10^{-3})^4} \text{ MPa} = 10.55 \text{ MPa}$$

$$\tau = \frac{F \cdot S_z^*}{I_z b} = \frac{-10 \times 10^3 \times 80 \times 40 \times 60}{\frac{1}{12} \times 10^{-3} \times 80 \times 80} = -0.88 \text{ MPa}$$



$$\sigma_{\alpha} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos 2\alpha - \tau_{xy} \sin 2\alpha$$

$$\leq \frac{\sigma_x + \sigma_y}{2} + \sqrt{\left(\frac{\sigma_x - \sigma_y}{4} \right)^2 + \tau_{xy}^2} = \frac{1}{2} \times 10.55 + \sqrt{\frac{10.55^2}{4} + 0.88^2}$$

$$\sigma_{\min} = \frac{\sigma_x + \sigma_y}{2} - \sqrt{\left(\frac{\sigma_x - \sigma_y}{4} \right)^2 + \tau_{xy}^2} = -0.06 \text{ MPa} \approx 10.66 \text{ MPa}$$

$$\tan 2\alpha_0 = \frac{2\tau_{xy}}{\sigma_x - \sigma_y} = \frac{-2 \times 0.88}{10.55} \therefore \alpha_0 \approx 4.75^\circ$$

18讲-应力圆-作业1

单元体如图所示。试利用应力圆的几何关系求：

- (1) 指定斜截面上的应力；
- (2) 主应力的数值；
- (3) 在单元体上绘出主平面的位置及主应力的方向

11, $\sigma_x = 0$ $\sigma_y = -20 \text{ MPa}$ $\tau_{xy} = -50 \text{ MPa}$.

两个点 $(0, -50)$ $(-20, 50)$

由图得 $C(40, 10)$

$\therefore \sigma_{45^\circ} = 40 \text{ MPa}$ $\tau_{45^\circ} = 10 \text{ MPa}$

(2) $R = \sqrt{50^2 + 10^2} = 51$

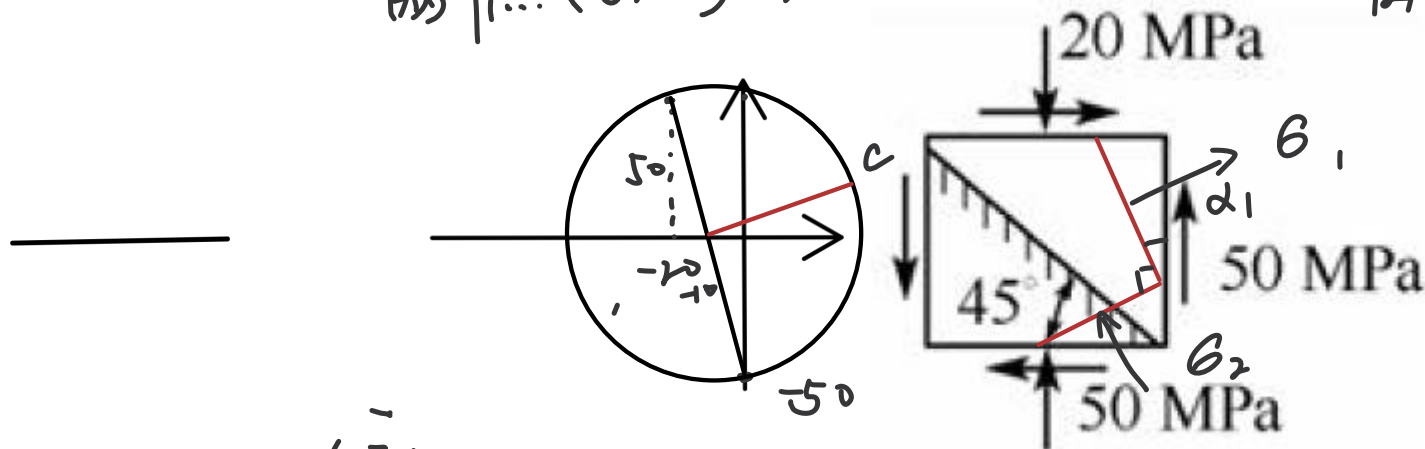
$\therefore \sigma_1 = 41 \text{ MPa}$

$\sigma_2 = -61 \text{ MPa}$

$\sigma_3 = 0$

$\alpha_1 \approx 39^\circ 35'$

$\alpha_2 = 50^\circ 25'$

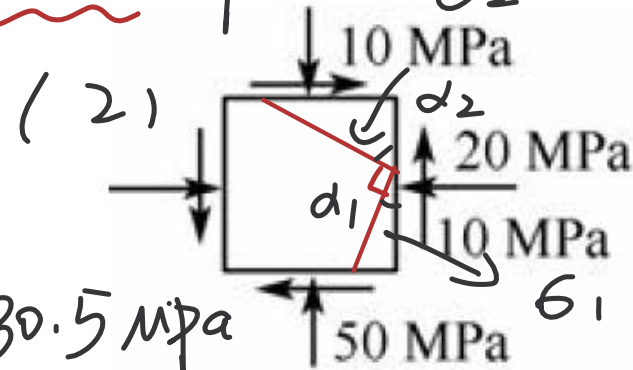
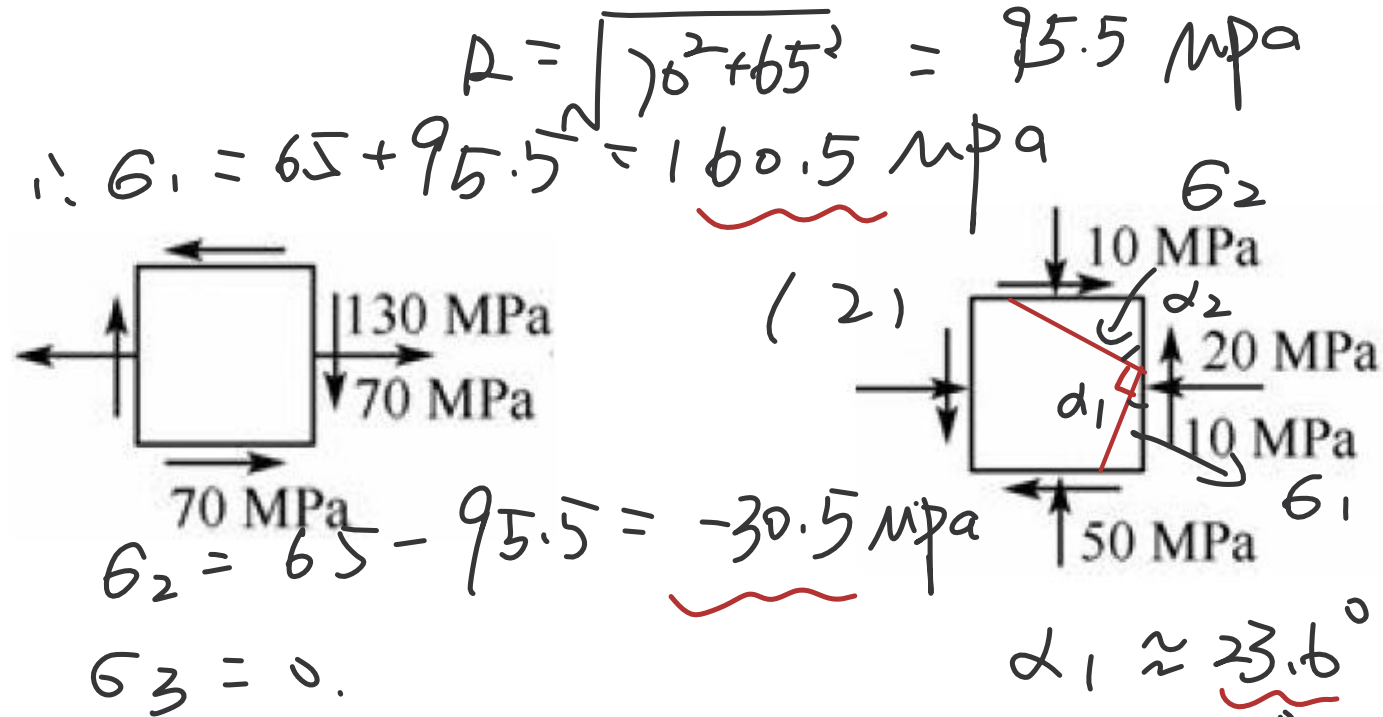


(3)

18讲-应力圆-作业2

对下图所示的每个单元体，试利用应力圆的几何关系求：

- (1) 主应力的数值；
- (2) 在单元体上绘出主平面的位置及主应力的方向。



$$\alpha_1 \approx \underline{23.6^\circ}$$

$$\alpha_2 \approx 90^\circ - 23.6^\circ = \underline{66.4^\circ}$$

1. 两个点, $A(130, 70)$
 $B(0, -70)$

